

A Network-Based, Multidisciplinary Approach to Intention Inference

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ABSTRACT

The ability to successfully infer private behavioral intentions using publicly-available digital records has a far-reaching impact. Unlike other private attributes such as demographic attributes, an intention oftentimes leads to a near-future behavior. Prior knowledge about such future behaviors can be seen as “actionable intelligence” and constitutes a significantly bigger risk to users’ privacy than knowledge of non-behavioral attributes. In this paper, we present a novel, multidisciplinary methodology for behavioral-intention inference. Using Bayesian-Networks, we model a behavioral intention using a set of causes that influence the intention’s formation, a set of effects that are caused by the intention, and various dependency relations within and between those sets. Unlike the methodologies used in prior attribute-inference works, which are oftentimes tailored to a single target attribute, our methods can be applied to different types of intentions from a diverse set of domains as we demonstrate by applying our model to multiple real-world intention-inference tasks.

CCS CONCEPTS

• **Human-centered computing** → *Social media*; • **Computing methodologies** → *Bayesian network models*.

KEYWORDS

Bayesian Networks, social networks, attribute inference

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1 INTRODUCTION

Knowledge of social-network users’ intentions has immense potential to improve the design of recommendation systems, ad-targeting mechanisms, public-health campaigns, and other social and commercial endeavors. At the same time, such knowledge can have a detrimental effect on users’ privacy. In this paper, we are interested

in inferring intentions of social-network users using public data extracted from their social-network profiles.

The paper makes the following contributions:

A unique focus on offline, time-varying behavioral intentions Unlike other existing works that tackle the task of private-attribute inference, ours is the first work that aims at inferring *offline and dynamic*, non-politically-related behavioral intentions of social-network users (*i.e.*, a *user-centric* approach) solely based on *public social-network data*. Other works either focus on online intentions or time-invariant preferences; use private data or non-social-network data; or take an “object-centric” approach by trying to infer the intention associated with a single, standalone and contextless social-network “object” such as a post or a tweet. Furthermore, some of the behavioral intentions that we consider in this paper, such as borrowing intentions and job-searching intentions, have never been studied in any prior machine learning (ML) or social-network-related work.

A new multidisciplinary methodology for the inference of behavioral attributes We introduce a novel, multidisciplinary methodology for the inference of behavioral attributes such as decisions and intentions. We design modular bayesian-network models that are able to capture the evolving nature of the human decision-making process by combining data and priors from multiple domains such as behavioral psychology, sociology, human-computer interaction (HCI), *etc.* Our methodology handles common challenges in social-network mining such as incomplete datasets, unlabeled data and bidirectional influence between features and the target variable.

2 RELATED WORK

Inference of private attributes using social-network data has been extensively researched. Inferring users’ personality type was investigated in [10, 11] using regression models and Twitter/Facebook data, respectively. Youyou *et al.* [38] showed that automatic inference methods that rely on Facebook likes achieve better prediction accuracy than those achieved by asking the users’ friends. Staiano *et al.* [36] used data gathered through smartphones such as calls and texts; their results significantly vary across different personality dimensions.

Demographic attributes’ inference is another well-studied topic, with age and gender being the most researched attributes [20, 33]. A related stream of research focuses on psychological and mental conditions. Depression is the most researched condition, followed by anxiety and stress [7, 31].

The common denominator of all the above works is that they focus on attributes that are either static (their values rarely change), non-self controlled, or both.

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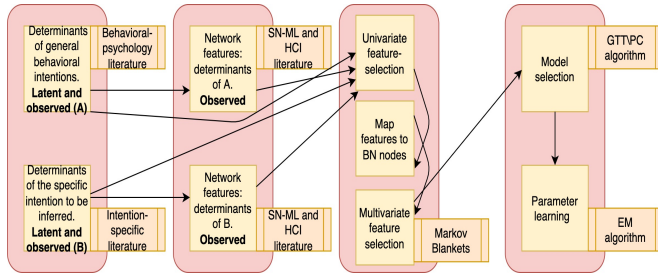


Figure 1: An illustration of our intention-inference methodology

Inference of self-controlled attributes has also been extensively studied. However, such works focus on the inference of opinions and attitudes rather than behavioral attributes [35, 40]. While a substantial amount of work does study different types of behavioral attributes, their goals are different than ours. Such works study general correlations between network or linguistic features and a given behavior, identify the prevalence of a certain behavior among the general population, or classify social-network textual objects such as tweets or posts. For example, while there exists a considerable amount of work about the use of social networks for monitoring public health, none of those works aims at inferring vaccination intent of a given social-network user at a given point in time. Rather, existing works analyze collective sentiment towards vaccinations [6, 25], track the spread of infectious diseases [32], or perform classification of stand-alone social-network objects according to vaccination attitudes or intentions of the object’s creator [15],[1].

Inferring time-varying, behavioral attributes using public social-network data has therefore been hardly researched, with two exceptions: voting intentions and *online* purchase intentions. There are several key differences between this work and prior ML work on PI. First, the majority of existing works examine general buying preferences rather than time-varying PIs [42]. Other works try to infer PI of stand-alone social-network objects (content-centric) rather than PI of social-network users (user-centric), an approach which is inherently biased [2, 12]. The remaining works that do try to infer a user-centric, time-varying PI use data derived solely from E-commerce platforms. Such data is both platform-specific and oftentimes considered private, unlike our use of public social-network data [28]. The closest work to ours is [22] which infers PI of Pinterest users using temporal features and a logistic regression model. However, they only consider online purchases and do not differentiate between different product categories.

3 DATA COLLECTION

We designed and distributed a comprehensive survey, created and hosted using Qualtrics survey platform. The first part of our survey contained questions about the participants’ personal attributes, as discussed in Section 5. The second part contained the following statements, which users were asked to rank (as well as dummy statements about unrelated intentions): “I am planning to start a weight-loss regime within the next 1-4 weeks” and “I am currently trying to lose weight” (weight-loss intentions); “I am planning to

look for a new job within the next 1-4 weeks” and “I am currently looking for a new job” (job-searching intentions); “I am planning to apply for a loan within the next 1-4 weeks” (borrowing intentions); “I received a flu vaccine this season” — depending on the answer to that statement the following statement was presented for either the upcoming (2020-2021) flu season or the next season (2021-2022): “I am planning to get vaccinated against influenza this upcoming fall-winter/next year” (vaccination intentions); “I am planning to make a travel-related purchase within the next 1-4 weeks” (travel-purchase intentions).

We implemented several methods for identifying and excluding data from participants who answered unreliably. First, we eliminated responses from participants who took the survey more than once. We took a conservative approach and discarded responses that came from the same IP address. Second, to ensure the eligibility of participants, they had to complete a screening questionnaire before taking the survey. The questionnaire was carefully designed, in order to prevent respondents from inferring the qualifications we were looking for and taking the survey without being an eligible participant. We avoided yes/no questions regarding the qualification needed and used multiway questions instead, as yes/no questions tend to insinuate the “correct” answer in order to pass the screening questionnaire. In addition, we disguised the real screening questions among other dummy questions. For example: instead of directly asking: “do you have a Facebook account”? we asked a series of identical multiway questions about news consumption habits from each of the media platforms. For example, the question “how often do you consume news via TV?” had the following five answers: More than 5 times a day; 3 times a day; once daily; Never; I do not have a TV. In the same question that dealt with Facebook, the last answer was replaced by “I do not have a Facebook account”. Third, we included control questions to ensure that the respondents were providing reliable data; those were fairly straightforward questions which asked the same question multiple times, in different parts of the survey, and using a slightly different terminology. We excluded participants who failed in one or more of the control questions. All survey data was anonymized after collection. We informed participants that their responses would be used for academic research.

Training and test datasets include data obtained from Amazon Mechanical Turk (MTurk), Facebook, Instagram and LinkedIn. 1300 respondents participated in our survey. For each intention, j , we randomly selected 780 records to serve as a training set for intention j ($\mathcal{D}_j^1$) and 520 records as a test set for intention j ($\mathcal{D}_j^2$). The size of our datasets is similar to datasets used in prior work in which labels are obtained using surveys or other means that require the users’ active participation. Examples include [16] (523), [5] (956), [19] (1216), [41] (400), [7] (1583), [11] (279), [36] (53).

4 METHODOLOGY

“Intentions are people’s decisions to perform particular actions” [34]. In this paper, we aim at understanding to what extent we can infer behavioral intentions of social-network users. In order to do that, we build a bayesian-network model that leans on intentions’ most influential factors as shown in behavioral psychology literature [9, 13, 34]. We split those factors into two groups: static factors, such as personality, demographic attributes and self-efficacy, and

dynamic factors such as emotions, interest and opinion. A significant challenge, however, is the fact that the values of some of those determinants (e.g. personality) can not be directly obtained from the user’s social-network profile (“latent variables”). Therefore, we enrich the model with various observed network features which may assist in both inferring the target intention and inferring each intention’s latent determinants. For instance, body image is known to have an important influence on weight-loss intentions [23]. However, most users do not publicly reveal their body image on social-network accounts. Therefore, we can use the following chain of dependency relations among 1st-2nd-3rd-layer features: (number of self-tagged photos, number of friends, public-figures-non-public-figures following ratio) $\leftarrow$ Body image $\rightarrow$ Weight-loss intentions.

In order to create our bayesian-network models we take the following steps: first, we identify the best set of determinants of general behavioral intentions using existing behavioral-psychology literature. Second, for each intention, we identify its unique determinants using existing literature which investigates that specific intention. Third, we identify the set of network features that are known to have a strong relation to the set of general and intention-specific determinants described above; the priors used for the third step, collected from existing literature, are not as strong as the priors used for stages 1 and 2 but are still informative — especially those collected from prior attribute-inference works and human-computer interaction (HCI) literature. Fourth, the final feature set of each intention is determined using priors, feature-selection methods, or both. Finally, the set of selected features is mapped into network nodes and the structure of the bayesian network is determined using a combination of model-selection algorithms and priors.

The diagram in Figure 1 illustrates our approach as detailed above. In Section 7, we present our models’ results when used for the inference of five behavioral intentions: weight-loss intentions (WI), vaccination intentions (VI), travel-purchase intentions (PI), borrowing intentions (BI) and job-searching intentions (JI).

Use of bayesian networks: The choice of bayesian networks for our inference tasks was motivated by several properties of bayesian networks that specifically suit the nature of our inference task and the nature of our datasets. We highlight two such properties.

Bayesian networks are able to work *directly* on incomplete datasets. Other popular discriminative classifiers can not work directly on datasets with missing values (missing values must be imputed). Since our datasets contain a large number of missing values which correspond to attributes that the user has not publicly revealed on one of her social-network accounts, choosing a model that can naturally handle missing values without the need for imputation is crucial for the model’s performance. Another important property of bayesian networks is their ability to incorporate prior information in the inference process. This fact is of special importance for our inference task as at least for some of the model’s dependency relations there exist very strong priors obtained from existing statistics (Census data) as well as existing behavioral-psychology, sociology and human-computer interaction literature. Such priors allow us to

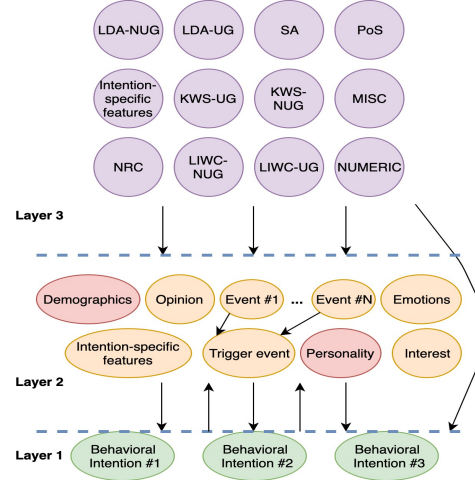


Figure 2: A diagram of our intention-inference bayesian-network model

work better on medium-sized datasets, noisy datasets and incomplete datasets as well as provide us strong reasons to believe that our models’ independence assumptions are correct.

5 FEATURES

A diagram of our intention-inference model is shown in Figure 2. Note that Figure 2 is brought for illustration purposes and thus only features edges between layers; edges between specific nodes must be determined separately for each intention according to its own unique priors, intention-specific features and results of feature selection methods applied to it. To avoid a large conditional probability table (cpt), we used a layering-divorcing technique and created a layered bayesian network model: The first layer contains the target intention nodes that we aim at inferring. The second layer contains either latent or partially-observed nodes which represent external and internal factors that are highly influential on the formation of behavioral intentions. The third layer contains observable network features. They serve two purposes: assisting in inferring the behavioral intentions, and serving as observed predictors for second layer’s latent variables.

5.1 Second-layer features

Values of second-layer features were obtained using our surveys. In order to simulate a real-world inference task (which only considers the public portion of online profiles), values of *latent* second-layer features were omitted from our test sets (treated as missing values); instead, we tried to infer them using *publicly available* network features.

Personality This variable represents five broad dimensions of personality obtained from the “Big Five” model of personality dimensions. The big five model distills personality into five traits: neuroticism, extraversion, agreeableness, conscientiousness, and openness to experience. To measure the Big Five personality traits among survey participants we used a short version of the Big Five Inventory based on BFI-10 [30].

Demographic attributes We considered the following demographic attributes: age, gender, ethnicity (and country of origin), marital status, occupation group, income (latent variable). Only a subset of those attributes was used in each model.

Situational variables Events that might trigger a certain behavioral intention. Those events include personal-life transitions, professional-life transitions, external events (such as a holiday or an election), *etc.* Priors were obtained for some intentions. For instance, life-transitions are shown to have an important impact on weight-loss intentions [3].

Emotions Different emotions may serve as either the cause of a behavioral intention or as its effect. Therefore, we went beyond the binary emotion-representation (positive-negative) and also considered fine-grained emotions. The most studied model of discrete emotions is the Ekman model [8] which posits the existence of six basic emotions: anger, disgust, fear, joy, sadness and surprise. Since momentary emotion ratings are not particularly indicative of the behavioral intentions explored in this work, survey participants were presented with eight emotion categories (six basic emotions and two positive-negative emotion categories) and were asked to rate their feelings over the past week/month/three months *in general*.

Interest, Opinion Those variables represent the user's level of interest and opinion regarding topics related to a given intention.

5.2 Network features

The value of a network feature was included in our datasets only if it was part of the public portion of one of the user's social-network profiles.

Numeric features (NUMERIC) We considered statistics about the user's activity (number of posts, status updates, number of uploaded photos *etc.*), reactions to the user's content (number of tagged photos, for instance) and the user's reactions to others' content. The latter measure was sparse, as both Facebook and Instagram limit the visibility of such reactions. We also considered basic statistics about the users' network, but we limit ourselves to statistics that are both publicly visible and can be directly extracted from the user's own social-network profile/s (number of friends, followers-following ratio, public-figures-non-public-figures following ratio, *etc.*).

Raw Textual features (TEXT) Textual features were classified as either user-generated (UG) features (including textual content that was written by the user), or non-user-generated (NUG) features (textual features that were not written by the user such as likes (Facebook) or hashtags (Instagram)). We limit ourselves to textual content that is both publicly visible and was either produced by the target user, or can be directly extracted from the user's own social-network profile/s. Raw textual features were not directly fed to our models. Instead, each textual feature was analyzed using various linguistic methods as described below; Textual-features-related-nodes in our network represent averaged discretized score or frequency of a given category of a given linguistic feature among the user's textual features. Such nodes represent the prevalence of a specific linguistic category among the entire set of raw textual features.

Miscellaneous features (MISC) Miscellaneous features include features that are neither numeric nor textual, such as the mere existence of various social-network accounts, visibility level/s that the

user has chosen to apply to her social-network accounts, profile attributes from which demographic attributes can be extracted, *etc.* MISC features can be seen as social-network accounts' metadata rather than data itself (NUMERIC, TEXT).

5.3 Linguistic features

We use a broad range of linguistic features, created based on our raw textual features.

Keyword-search (KWS-UG, KWS-NUG) For a given intention, or an event, $\mathcal{A}$, we manually identified the most prominent keywords related to $\mathcal{A}$. We then performed a keyword search on our textual features. For some textual features, items that were found to contain a relevant keyword were further processed using a second method, depending on the nature of $\mathcal{A}$ and the nature of the textual feature. This resulted in two groups of features, KWS-UG and KWC-NUG (keyword search applied to user-generated/non-user-generated content).

LIWC (LIWC-UG, LIWC-NUG) LIWC is a text analysis tool that is widely used in psychological studies [37]. Each list of words is associated with a semantic or syntactic category, such as negations, adverbs or tone. LIWC analysis was applied to UG and NUG textual features. However, with regards to NUG features, we only considered a small subset of LIWC categories, mostly concerning topics and sentiment.

Sentiment analysis, part-of-speech tagging (SA, PoS) These were only applied to KWS-UG (SA and PoS) and KWS-NUG (SA), *i.e.*, items that were found to contain at least one relevant keyword. SA was applied to items that were found to contain keywords that relate to the behavioral intention to be inferred, in order to assess the user's opinion on related topics. The use of PoS tagging was more implicit; it was applied to items that were found to contain keywords that relate to events in order to assess whether an event is relevant to each inference task (use of first-person writing, tensed verbs *etc.*).

Topic modeling (LDA-UG, LDA-NUG) Topics were extracted using Latent Dirichlet Allocation (LDA). Shorter features (such as likes) and longer features (such as posts) were considered separately using different parameters. As many likes only contain names (*e.g.* brand names), we considered a like to be both the like's title as well as the category to which it belongs.

Emotions (NRC, LIWC) We automatically quantify emotions from our UG textual features using LIWC and NRC. NRC is a publicly available lexicon of words associated with different emotions, as well as general positive and negative sentiment [26]. We assign a predicted emotion to each UG textual feature and then average across all users' features.

6 MODEL SELECTION

The features described in Section 5 were first fed into our feature selection algorithm. Only features that were found to be relevant for the inference of a given intention were included in the intention's bayesian-network model.

We designed a two-level, hybrid feature-selection method. Due to the high number of correlations between different features, we opted for a bayesian-network-based-feature-selection method

rather than a univariate filter-based approach. However, performing feature selection using only a bayesian network may lead to overfitting. In addition, the vast majority of bayesian-network-learning algorithms require complete datasets. To combine the best of both worlds, we employed a hybrid feature selection approach. First, a simple, univariate feature selection method was applied to a subset of the features on which we didn't have strong prior information. For that purpose, we used a mutual information-based feature selection method and removed all the features that received a score below a certain threshold. Mutual information measures the dependency between variables and is thus highly suited for bayesian-network-based models. The resulting features, as well as the set of latent and high-prior features were the input for the second feature-selection phase which utilizes a wrapper feature-selection method using bayesian networks.

Model selection was performed using a combination of automatic structure-learning algorithms (Greedy-Thick-Thinning [4]), and priors obtained from existing behavioral-psychology literature, intention-specific literature and Census data, such as [9, 14, 17, 18, 23, 29]. The balance between automatic structure-learning algorithms and the use of priors for structure elicitation, as well as the initial parameters for the structure-learning algorithms (when applicable) were validated using cross-validation. Note that while information gathered from prior literature would have probably been sufficient to model most of the meaningful dependency relations between an intention and second-layer features, relations between third-layer features and other features, as well as between third-layer features and the target intentions can not be captured solely using priors, as those types of relations are not as extensively studied as behavioral intentions-second-layer features relations.

Nodes in our bayesian networks represent either discretized values of numeric features; categorical features such as demographic attributes; or averaged discretized frequency of a given category of a given linguistic feature among the user's textual features. Such nodes are essentially aggregators as their value represents the prevalence of a specific linguistic category among the entire set of the user's raw textual features. Finally, situational variables are connected to the target node through an intermediate node, "trigger event".

Table 1: Results of the bayesian-network models presented in this paper

Intention	% intending	Micro F1	Macro F1
Vaccination intentions	.58	.732	.73
Weight-loss intentions	.38	.832	.815
Borrowing intentions	.17	.663	.54
Travel-purchase intentions	.24	.763	.691
Job-searching intentions	.19	.704	.623

7 EXPERIMENTAL RESULTS

For a given intention, j , we trained its bayesian network, BN_j , using $\mathcal{D}_j^1$ and tested it on $\mathcal{D}_j^2$. For that purpose, we used the Lauritzen-Spiegelhalter junction tree algorithm [21], which was applied to the original, incomplete datasets. Note that some of our datasets are highly imbalanced. Moreover, in order to simulate a real-world inference task we only consider the public portion of each user's online social-network profiles. Therefore, each dataset contains a large number of missing values which correspond to attributes that the user has not publicly revealed on her social-network accounts. Those facts make the inference task highly challenging, both as a stand-alone task and compared to inference tasks in prior attribute-inference works. Table 1 provides a summary of our results. We report Micro F1 and Macro F1 scores. It can be seen that the "intending" class is, almost always, the minority class. As a baseline, we chose a Multinomial Naive Bayes (MNB) classifier.

As can be seen, different intentions achieved significantly different Micro F1 and Macro F1 scores. BI's score is the lowest, whereas WI's score is the highest. A possible explanation for BI's performance is that applying for a loan is an intention that is oftentimes not publicly shared in social networks. However, other non-publicly shared intentions such as JI scored significantly better than BI. This can be attributed to the fact that we were able to find other strong predictors for JI which don't depend on user-generated content, whereas for BI we failed to do so.

As seen in Figure 3, our model outperforms a MNB classifier on all five intentions. The differences between the two classifiers varies between intentions. A possible explanation for those differences is the varying number of dependencies among each model's variables, or the existence of dependencies that are of specific importance for the inference task.

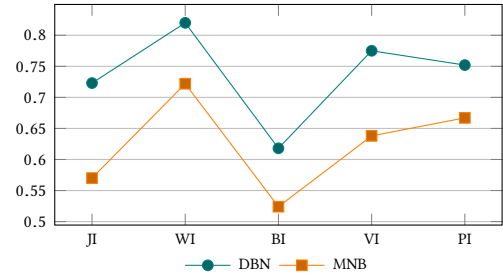


Figure 3: Average ROC AUC scores

Though the set of most determinant features vary between intentions, some features appeared highly influential on majority of intentions. Those include KWS features (.078 average sensitivity score), intention-specific features (.081), and MISC features (.09); the latter is important as MISC features are usually much easier to collect and process than numeric or textual features, yet are oftentimes ignored in attribute-inference works. Emotions, on the other hand, were found to have very little influence on all intentions. This finding contradicts existing strong priors [24, 27, 39] which find emotions to be an important determinant of some intentions (mostly short-termed ones). A possible explanation is the fact that

emotions change rapidly, and therefore we were not able to correctly capture and model them with static models. Surprisingly, LDA features were also among the features with lowest influence with respect to all intentions (though were moderately influential on personality-traits nodes). This can be attributed to our use of keyword features, which can be seen as "targeted" LDA features with predefined topics that are known to be highly related to each intention.

8 FUTURE WORK

In this work, we have treated intentions as time-invariant attributes by modeling each intention using a static bayesian network. Decision-making, however, is a dynamic process. Thus, in future work we will explore the extent to which temporal models might improve the inference task. Such an approach can be particularly useful for those behavioral intentions with a relatively short intention-behavior interval; in such situations, not only the absolute values of network features but also the *differences, or changes* between values of network features in different points in time may significantly improve the predictive power of the model.

Due to the nature of our inference task and our social-network datasets, a natural choice of a temporal model is a dynamic bayesian network that shares most of the advantages of bayesian networks, yet extends them to include temporal relations between different variables and each behavioral intention. Dynamic bayesian networks, when used with survey data that is collected in multiple waves with a lag period that matches the unique intention-behavior interval of each intention, have the potential to improve both inference accuracy of current intentions and prediction accuracy of future intentions.

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